

CS 270 – Lab 9 Q4

Question 4: (M x L)

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;input-contract: x is a nonnegative integer,
;                and L is a nonempty list of positive integers.
;output-contract: (M x L) is a recursively defined integer
;                -- pattern redacted for students
(define (M x L)
  (if (zero? x) (first L) (M (- x 1) (rest L)))))

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Use Equational Reasoning to prove (M 1 '(8 4 7)) evaluates to 4.

Note: this proof should fill all 10 lines

Expression	Justification
(M 1 '(8 4 7))	Premise
(if (zero? 1) (first '(8 4 7)) (M (- 1 1) (rest '(8 4 7))))	Apply definition of M with x=1 and L='(8 4 7)
(if #f (first '(8 4 7)) (M (- 1 1) (rest '(8 4 7))))	Evaluate zero?
(M (- 1 1) (rest '(8 4 7)))	Evaluate if
(M 0 (rest '(8 4 7)))	Evaluate -
(M 0 '(4 7))	Evaluate rest
(if (zero? 0) (first '(4 7)) (M (- 0 1) (rest '(4 7))))	Apply definition of M with x=0 and L='(4 7)
(if #t (first '(4 7)) (M (- 0 1) (rest '(4 7))))	Evaluate zero?
(first '(4 7))	Evaluate if
4	Evaluate first